

Lecture#12

Normal Distribution

A random variable X is said to have a normal distribution with mean μ and variance σ^2 ($-\infty < \mu < \infty$ and $\sigma^2 > 0$) if X has a continuous distribution for which the probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

*Normal probability law:

$$P(\mu-3\sigma < X < \mu+3\sigma) = 99.73\%$$

$$P(\mu-2\sigma < X < \mu+2\sigma) = 95.45\%$$

$$P(\mu-\sigma < X < \mu+\sigma) = 68.27\%$$

Note. Normal distribution is also called Gaussian distribution.

Standard Normal Distribution:

If a random variable X has a normal distribution with mean μ and variance σ^2 (i.e. $X \sim N(\mu, \sigma^2)$), then the variable $Z = \frac{X-\mu}{\sigma}$ will be called a standard normal variable (or Z score) and its distribution is referred to as the standard normal distribution having the following density function:

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{Z^2}{2}}, \quad -\infty < Z < \infty$$

$$\begin{aligned} E(Z) &= E\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma} E(X - \mu) = \frac{1}{\sigma} [E(X) - E(\mu)] \\ &= \frac{1}{\sigma} [\mu - \mu] \\ &= 0 \end{aligned}$$

$$\begin{aligned} V(X) &= V\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma^2} v(X - \mu) = \frac{1}{\sigma^2} [V(X) + V(\mu) - 2Cov[X]] \\ &= \frac{1}{\sigma^2} [\sigma^2 + 0 - 0] \\ &= 1 \end{aligned}$$

Thus, $Z \sim N(0,1)$

The cumulative distribution function (cdf) of the standard normal variable Z is usually denoted by $\phi(Z)$. Thus

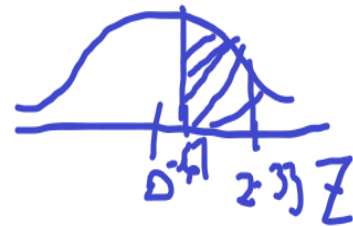
$$\phi(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^Z e^{-\frac{Z^2}{2}} dZ$$

Example: The GPA score of 80 students of the Department of Physics of University of Dhaka in their 1st year final exam was found to follow approximately a normal distribution with mean 2.1 and standard deviation 0.6. How many of these students are expected to have a score between 2.5 and 3.5?

Solution: Let X be the GPA score of the 80 students.

$$X \sim N(2.1, 0.6^2)$$

$$\begin{aligned} \text{Now, } P(2.5 < X < 3.5) &= P\left(\frac{2.5-2.1}{0.6} < \frac{X-\mu}{\sigma} < \frac{3.5-2.1}{0.6}\right) \\ &= P(0.67 < Z < 2.33) \end{aligned}$$



$$\begin{aligned} &= P(Z < 2.33) - P(Z < 0.67) \\ &= 0.9901 - 0.7480 = 0.2421 \end{aligned}$$

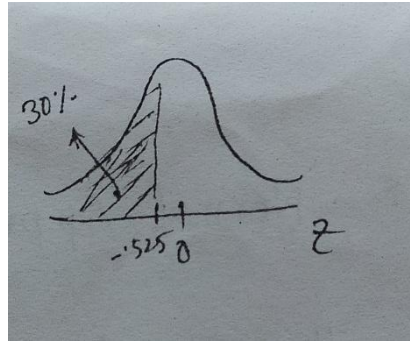
Hence, 24.21% or approximately $0.2421 \times 80 = 20$ students out of 80 are expected to make a score between 2.5 and 3.5.

Example: If X is a normal variate with mean 25 and variance 9, find K such that

- i. 30% of the area under the normal curve lies to the left of the distribution.
- ii. 15% of the area under the normal curve lies to the right of the distribution.

Solution:

$$X \sim N(25, 9)$$



i.e., $\mu = 25, \sigma = 3$

$$1. \quad P(X < K) = 0.30 \Rightarrow P\left(\frac{X - \mu}{\sigma} < \frac{K - 25}{3}\right) = 0.30 \Rightarrow P\left(Z < \frac{K - 25}{3}\right) = 0.30$$

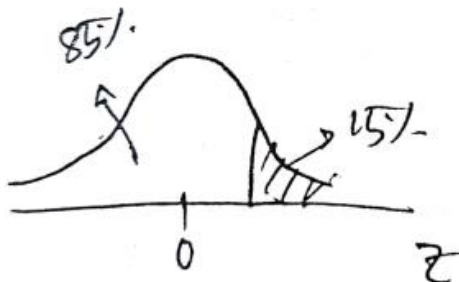
The standard normal Table shows that

$$P(Z < -0.525) = 0.30$$

Hence,

$$\begin{aligned} (K - 25)/3 &= -0.525 \\ \Rightarrow K &= 23.425 \end{aligned}$$

$$\text{ii. } P(X > K) = 0.15$$



$$\text{or, } P\left(Z > \frac{K - 25}{3}\right) = 0.15$$

$$\text{or, } P\left(Z < \frac{K - 25}{3}\right) = 1 - 0.15 = 0.85$$

The standard normal Table shows that

$$P(Z < 1.035) = 0.85$$

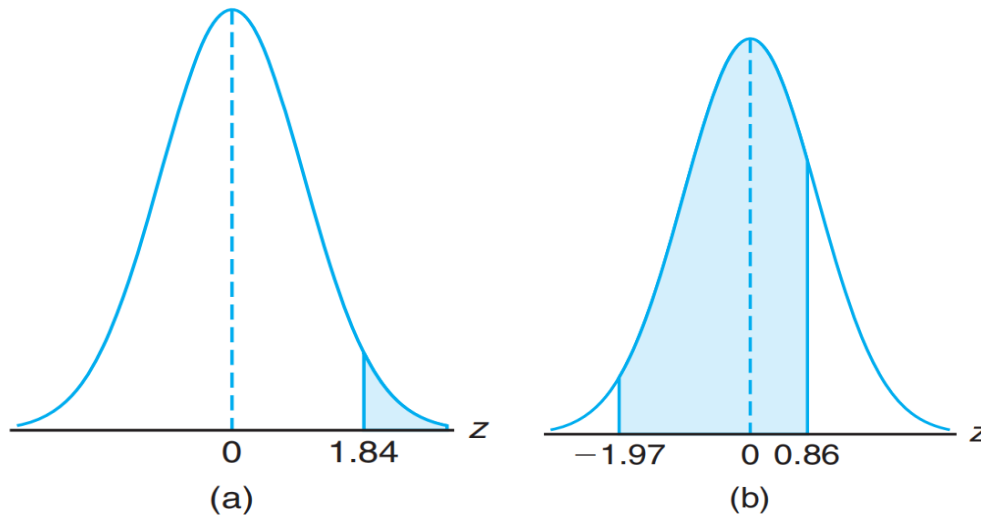
Hence,

$$\begin{aligned} (K - 25)/3 &= 1.035 \\ \Rightarrow K &= 28.105. \end{aligned}$$

Example: Given a standard normal distribution, find the area under the curve that lies

(a) to the right of $z = 1.84$

(b) between $z = -1.97$ and $z = 0.86$.

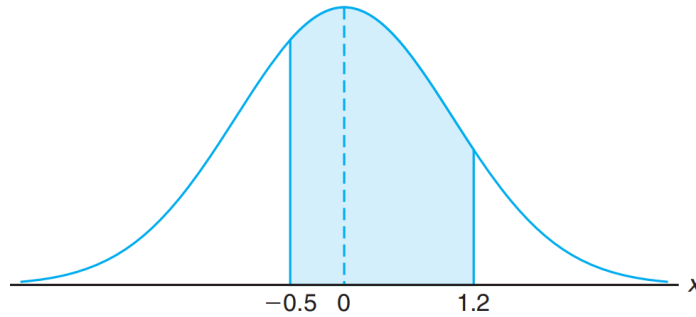


Solution: See Figure 6.9 for the specific areas.

(a) The area in Figure 6.9(a) to the right of $z = 1.84$ is equal to 1 minus the area in Z-table to the left of $z = 1.84$, namely, $1 - 0.9671 = 0.0329$.

(b) The area in Figure 6.9(b) between $z = -1.97$ and $z = 0.86$ is equal to the area to the left of $z = 0.86$ minus the area to the left of $z = -1.97$. From Z-table we find the desired area to be $0.8051 - 0.0244 = 0.7807$.

Example: Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$, find the probability that X assumes a value between 45 and 62.



Solution: The z values corresponding to $X_1 = 45$ and $X = 62$ are

$$Z_1 = \frac{45-50}{10} = -0.5 \text{ and } Z_2 = \frac{62-50}{10} = 1.2$$

Therefore,

$$P(45 < X < 62) = P(-0.5 < Z < 1.2).$$

$P(-0.5 < Z < 1.2)$ is shown by the area of the shaded region in the Figure.

This area may be found by subtracting the area to the left of the ordinate $z = -0.5$ from the entire area to the left of $z = 1.2$. Using Z-table, we have

$$\begin{aligned} P(45 < X < 62) &= P(-0.5 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.5) \\ &= 0.8849 - 0.3085 = 0.5764. \end{aligned}$$

Example: In an industrial process, the diameter of a ball bearing is an important measurement. The buyer sets specifications for the diameter to be 3.0 ± 0.01 cm. The implication is that no part falling outside these specifications will be accepted. It is known that in the process the diameter of a ball bearing has a normal distribution with mean $\mu = 3.0$ and standard deviation $\sigma = 0.005$. On average, how many manufactured ball bearings will be scrapped?

Solution: The distribution of diameters is illustrated by Figure 6.16. The values corresponding to the specification limits are $x_1 = 2.99$ and $x_2 = 3.01$. The corresponding z values are

$$Z_1 = \frac{2.99 - 3.0}{0.005} = -2 \text{ and } Z_2 = \frac{3.01 - 3.0}{0.005} = +2.0$$

Hence,

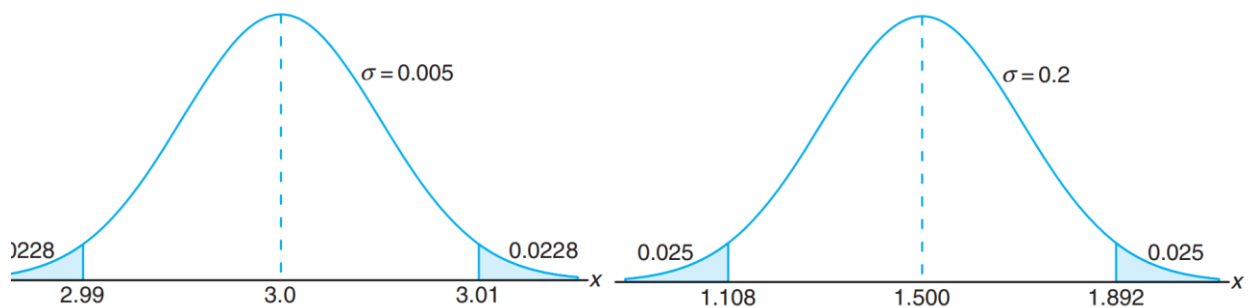
$$P(2.99 < X < 3.01) = P(-2.0 < Z < 2.0).$$

From Table A.3, $P(Z < -2.0) = 0.0228$.

Due to symmetry of the normal distribution, we find that

$$P(Z < -2.0) + P(Z > 2.0) = 2(0.0228) = 0.0456.$$

As a result, it is anticipated that, on average, 4.56% of manufactured ball bearings will be scrapped.



Continuity Correction

$$P(a \leq X \leq b) \approx P(a - 0.5 < X' < b + 0.5)$$

where X' is the constant normal variable

Let, $X \sim \text{Bin}(n = 10, P = 0.5)$. Find $P(2 \leq X \leq 4)$ using both binomial distribution and normal distribution:

Solution: Using binomial distribution:

$$\begin{aligned} &P(2 \leq X \leq 4) \\ &= \sum_x \binom{n}{x} P^x (1 - P)^{n-x} \end{aligned}$$

$$= \sum_{x=2}^4 \binom{10}{x} (0.5)^x (1 - 0.5)^{10-x} = 0.3662$$

Now, $P(2 \leq X \leq 4) \approx P(2-0.5 \leq X' \leq 4+0.5) = P(1.5 \leq X' \leq 4.5)$

Using normal distribution, $\mu = np = 5$, $\sigma^2 = np(1-p) = 2.5$

$P(1.5 \leq X' \leq 4.5)$

$$= P\left(\frac{1.5-5}{\sqrt{2.5}} \leq Z \leq \frac{4.5-5}{\sqrt{2.5}}\right)$$

$$= P(-2.21 \leq Z \leq -0.316)$$

$$= P(Z \leq -0.316) - P(Z \leq -2.21)$$

$$= 0.3764 - 0.0136$$

$$= 0.362.$$

Exercise.

1. The height of adult men in a city is normally distributed with a mean of 170 cm and standard deviation of 6 cm. Find the probability that a randomly selected man is taller than 180 cm.
2. Test scores are normally distributed with mean 70 and standard deviation 10. Find the probability that a student scores between 60 and 85.
3. Weights of newborns are normally distributed with mean 3.2 kg and standard deviation 0.4 kg. Find the weight that separates the lightest 10% of babies from the rest.
4. A machine fills bottles with a normally distributed volume of liquid, mean 500 ml and SD 8 ml. Probability that a bottle has more than 520 ml.
5. The lifetime of a type of bulb is normally distributed with mean 1200 hours and SD 100 hours. Probability that a bulb lasts less than 1050 hours or more than 1350 hours.
6. Given the normally distributed random variable x with mean 18 and standard deviation 2.5, find the value of k such that (i) $P(x < k) = 0.1539$, (ii) $P(x > k) = 0.1539$.

7. The effective life of a component used in a jet-turbine aircraft engine is a random variable with mean 5000 hours and standard deviation 40 hours. The distribution of effective life is fairly close to a normal distribution. The engine manufacturer introduces an improvement into the manufacturing process for this component that increases the mean life to 5050 hours and decreases the standard deviation to 30 hours. Suppose that a random sample of $n_1 = 16$ components is selected from the "old" process and a random sample of $n_2 = 25$ components is selected from the "improved" process. What is the probability that the difference in the two sample means $(\bar{x}_2 - \bar{x}_1)$ is at least 25 hours? Assume that the old and improved processes can be regarded as independent population.

8. The tensile strength of a certain metal component is normally distributed with a mean of 10,000 kilograms per square centimeter and a standard deviation of 100 kilograms per square centimeter. Measurements are recorded to the nearest 50 kilograms per square centimeter. Calculate the following cases and show the regions graphically
 - (i) What proportion of these components exceed 10,150 kilograms per square centimeter in tensile strength?
 - (ii) If specifications require that all components have tensile strength between 9800 and 10,200 kilograms per square centimeter inclusive, what proportion of pieces would we expect to scrap?

Binomial to normal approximation

Step 1: Conditions for approximation

For $X \sim \text{Binomial}(n, p)$, we can approximate X by a normal distribution $Y \sim N(\mu, \sigma^2)$ if:

1. n is large
2. Both $np \geq 5$ and $n(1 - p) \geq 5$

Then,

Mean, $\mu = np$, and variance, $\sigma^2 = np(1 - p)$

Step 2: Continuity correction

- Binomial is discrete; normal is continuous.
- To approximate $P(X \leq k)$, use $P(Y \leq k + 0.5)$
- To approximate $P(X \geq k)$, use $P(Y \geq k - 0.5)$

This is the **continuity correction**.

Step 3: Convert to Z-score

$$Z = \frac{X - \mu}{\sigma}$$

Then use the **standard normal table** to find probabilities.

Problem 1. According to Chemical Engineering Progress, approximately 30% of all pipework failures in chemical plants are caused by operation error.

- (i) What is the probability that out of the next 20 pipework failures at least 10 are due to operator error?
- (ii) What is the probability that no more than 4 out of 20 such failures are due to operator error? of the random sample of 20 such
- (iii) Suppose, for a particular plant, that out failures, exactly 5 are due to operator error. Do you feel that the 30% failures stated above applies to this plant? Comment.

Solution:

Given, Probability of operator error: $p = 0.30$, Number of failures sampled: $n = 20$, X = number of failures due to operator error.

$$X \sim \text{Binomial } (n = 20, p = 0.3)$$

Binomial pmf:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}; x = 0, 1, 2, \dots, n$$

- (i) **Probability that at least 10 are due to operator error**

$$P(X \geq 10) = \sum_{x=10}^{20} \binom{20}{x} p^x (1-p)^{20-x}$$

Check if we can use Normal approximation

- $n = 20, p = 0.3$. So, $np = 20 * 0.3 = 6$
- $n(1 - p) = 20 * 0.7 = 14$
- Both > 5 , so Normal approximation is reasonable.

Normal approximation

- Mean: $\mu = np = 6$, Standard deviation: $\sigma = \sqrt{np(1-p)} = \sqrt{20 * 0.3 * 0.7} = 2.049$
- Continuity correction: $P(X \geq 10) \approx P(Y \geq 9.5)$

$$\begin{aligned} \text{Now, } P(Y \geq 9.5) &= P\left(\frac{Y-\mu}{\sigma} \geq \frac{9.5-\mu}{\sigma}\right) = P\left(Z \geq \frac{9.5-6}{2.049}\right) = P(Z \geq 1.708) \\ &= 1 - P(Z < 1.708) = 1 - 0.956 = 0.044 \text{ (Approx.)} \end{aligned}$$

Answer (i): Approximately **4.4%**

(ii) Probability that no more than 4 are due to operator error

$$\begin{aligned} P(X \leq 4) &\approx P(Y \leq 4.5) = P\left(\frac{Y-\mu}{\sigma} \leq \frac{4.5-\mu}{\sigma}\right) = P\left(Z \leq \frac{4.5-6}{2.049}\right) \\ &= P(Z \leq -0.732) \end{aligned}$$

$$= 1 - P(Z \leq 0.732) = 1 - 0.767 = 0.233 \text{ (Approx.)}$$

Answer (ii): Approximately **23.3%**

(iii) Comment if exactly 5 out of 20 failures are due to operator error

$$\text{Here, } Z = \frac{X-\mu}{\sigma} = \frac{5-6}{2.049} = -0.49$$

Now, we calculate, $P(Z \leq -0.49) = 0.31 = 31\%$.

Comment: There is no strong evidence to reject the 30% operator-error rate. Observing exactly 5 failures due to operator error is consistent with the stated 30% rate.

$$\text{Or, Here, } \hat{p} = \frac{X}{n} = \frac{5}{20} = 0.25$$

Null hypothesis, $H_0: p = 0.30$ or $\mu = 6$

Alternative hypothesis, $H_1: p \neq 0.30$, or $\mu \neq 6$

$$\text{Test statistic, } Z_{cal} = \frac{\hat{p} - p}{\sigma_{\hat{p}}}; \text{ or, } Z_{cal} = \frac{X - \mu}{\sigma} \frac{5 - 6}{2.049} = -0.49.$$

Critical value at 5% level of significance, $|Z_{tab}| = 1.96$

As, $|Z_{cal}| < |Z_{tab}| \Rightarrow$ Not reject $H_0: p = 0.30$ or $\mu = 6$

Comment: There is no strong evidence to reject the 30% operator-error rate. Observing exactly 5 failures due to operator error is consistent with the stated 30% rate.